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Second Semester, 2026
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MAST20030: Differential Equations
Tutorial: Week 9

Question 1

Solve the following BVPs:

(a) $y'' + 2y' + y = 0$ where $y(0) = 1$ and $y'(2) = 0$,

(b) $y'' + 2y' + y = 0$ where $y(0) = 0$ and $y'(1) = 1$,

(c) $y''(x) + y(x) = x$ where $y'(0) = 1$ and $y'(\pi) = 1$.

Question 2

Compute the eigenvalues and eigenfunctions of the differential operator $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$, that satisfies the following boundary conditions:

$$f'(0) = f(\pi) = 0.$$

Question 3

Consider the heat equation

$$\frac{\partial \phi}{\partial t} = 4\frac{\partial^2 \phi}{\partial x^2}$$

for $0 < x < \pi$ and $t > 0$, subject to the boundary and initial conditions

$$\frac{\partial \phi}{\partial x}(0, t) = 0$$
$$\phi(\pi, t) = 0$$
$$\phi(x, 0) = \cos\frac{5x}{2}$$

(a) Show that separation of variables leads to the eigenvalue problem in Q2(a).

(b) Determine the solutions involving t corresponding to each eigenvalue, hence determine the general solution for ϕ that satisfies the boundary conditions.

(c) Apply the initial condition to determine the solution to the complete initial-boundary value problem.

(d) Verify that your complete solution satisfies the PDE, boundary conditions and initial conditions.

Question 4

[(Hard)]

(a) Let's revisit Question 2 but changing the boundary conditions to

$$f(0) = f'(0) \qquad \text{and} \qquad f(\pi) = f'(\pi),$$

show that there is an infinite number of positive eigenvalues and well as a single negative eigenvalue. What are the corresponding eigenfunctions?

(b) Find the general solution to the PDE in Question 3 but with the boundary conditions

$$\phi(0, t) = \frac{\partial \phi}{\partial x}(0, t)$$
$$\phi(\pi, t) = \frac{\partial \phi}{\partial x}(\pi, t)$$

What would you need to do to determine the series coefficients if you were given an initial condition?